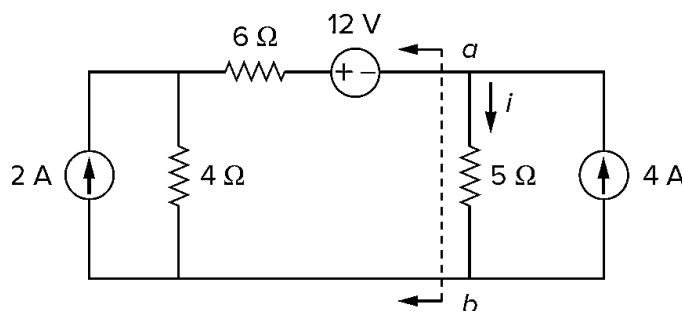


AELL101: Introduction to Electrical Engineering

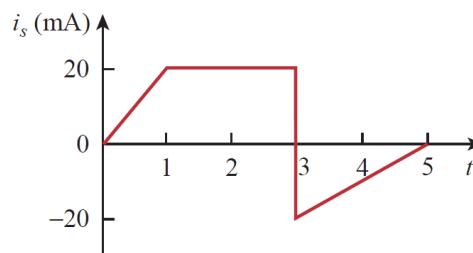
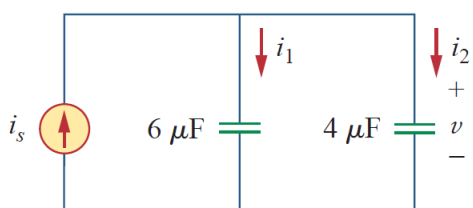
Midsemester Examination (100 marks; 2 hours)

Instructions: Write and sketch neatly. Merely stating the final numerical answer without specifying intervening steps, would fetch no marks. Please mention appropriate units against each answer. **No clarification will be provided. Make assumptions if you think some information is missing.** The answers to all parts of a question must be collated at the same place, starting on a fresh page.

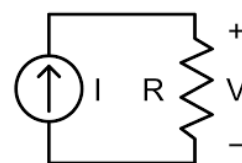
1. (10 marks) Consider the circuit in the figure shown on the right side.
 - (a) Obtain the Norton equivalent of the circuit to the left of terminals a-b.
 - (b) Use the result of part (a) to find current i .



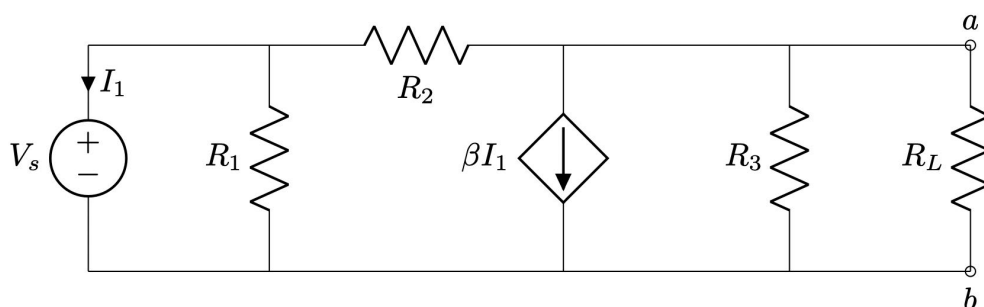
2. (10 marks) If $v(0) = 0$, find $v(t)$ and $i_1(t)$ in the circuit depicted below where the current source $i_s(t)$ as a function of time is also plotted below.



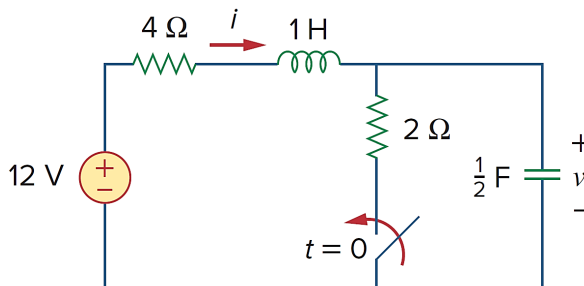
3. (10 marks) A current source in a circuit, shown in the right, has $I = 8 \cos(500\pi t - 25^\circ)$ A. Find
 - a. RMS value of current
 - b. Frequency of the current in Hz
 - c. Express I as phasor
 - d. Among voltage V and current I phasors, which one is leading?



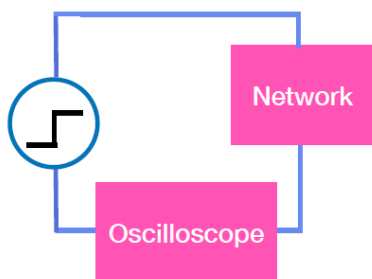
4. (15 marks) Determine the value of load resistance R_L for maximum power transfer in the circuit given below. Given parameters: $R_1 = 150 \text{ k}\Omega$, $R_2 = 1 \text{ k}\Omega$, $R_3 = 1 \text{ k}\Omega$, $\beta = 98$



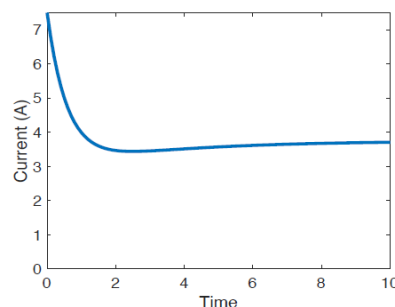
5. (15 marks) Find the complete response $v(t)$ and $i(t)$ for $t > 0$ assuming the switch was opened for a long time before it was closed at $t = 0$. Is the response overdamped or underdamped and why?



6. (15 marks) A black box of a circuit consisting of four circuit elements is given. The **four circuit elements are two equal resistors, one inductor and one capacitor**. All elements are fully discharged in the beginning. To find the configuration of this circuit, a step voltage of 15 V is applied at $t = 0$ and the current through an oscilloscope is measured, as shown below in fig (a). The measured current is plotted as function of time below in fig (b). **Find the black box circuit and give proper reasoning.** You need to find only connection diagram (series, parallel or some other combination) of all 4 elements, not the values.



(a)



(b)

7. (15 marks) Design a light that flashes automatically 2 times per second. You have following 4 electrical elements available to you along with wiring and other construction material:

(a) 60 V battery, (b) $10 \mu\text{F}$ capacitor, (c) resistance and (d) a neon lamp that remains non-conducting until the applied voltage across it reaches 30 V and then it instantly becomes an ideal conductor till the voltage drops back to 0 V. Once voltage reaches 0, it again becomes ideal insulator.

Device a simple RC circuit that will provide automatic flashing and specify the value of R. Explain your reasoning.

8. (10 marks) An engineering student is designing an electric automobile for a cross-country race where reliability (quick response to driver's command) is crucial. Whenever the switch S in the motor field circuit needs to be opened for the purpose of making current zero, she observes an arc across the contact. She reasons that arc can be eliminated by placing a capacitor across the switch as shown in the figure below. (a) Justify her reasoning (b) Considering the effect of C on the circuit, recommend an optimum value for C.

